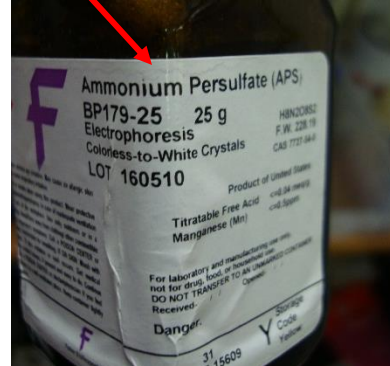


Long Gel Set-Up and Protocol

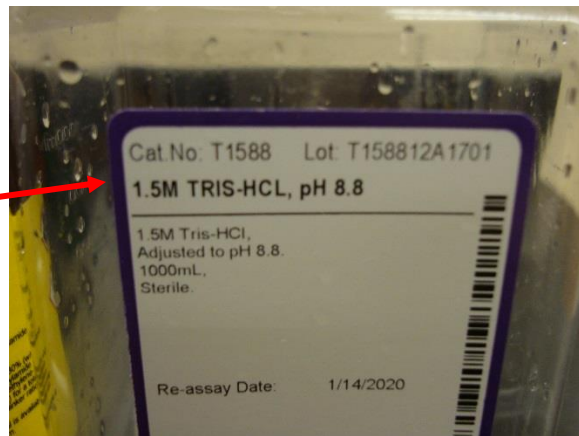


Work area where long gels are made

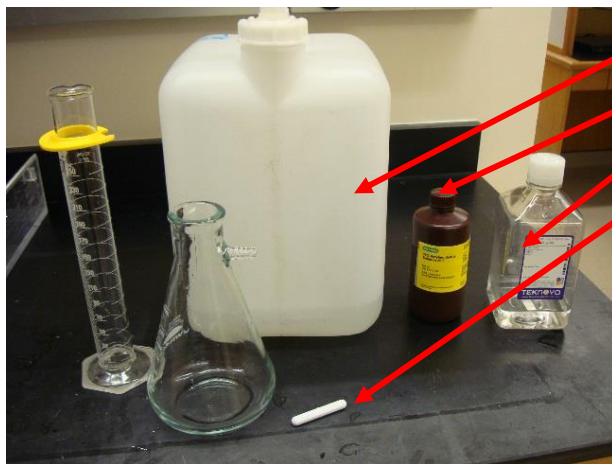
1. Set out 30% Acrylamide/Bis Solution (29:1), TEMED, and Ammonium Persulfate (APS) to warm at room temperature. (Found in the 4°C refrigerator)



2. Set out 1.5M Tris (pH 8.8). (Found lower right cabinet drawer where long gels are made)

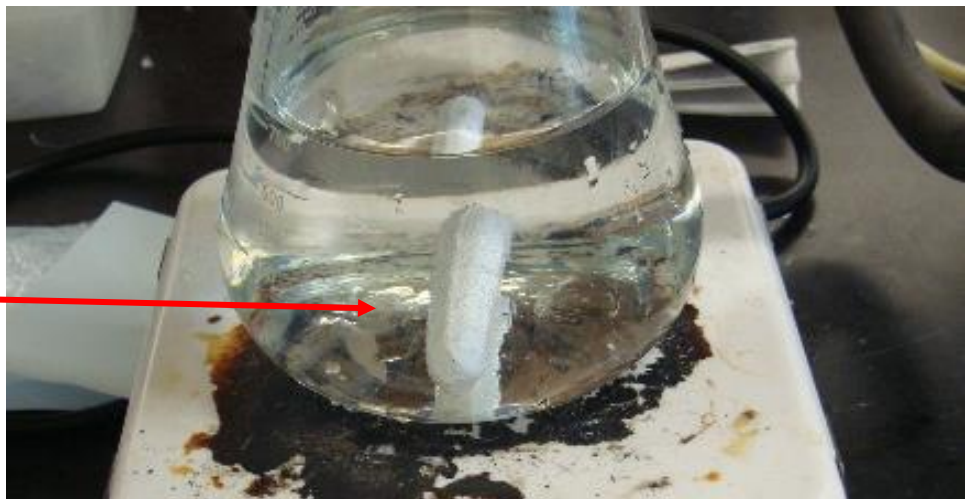


3. Make resolving gel by placing reagents into Erlenmeyer



- add 290mL ddH₂O
- 240mL 30% acrylamide solution
- 180 mL of 1.5M Tris (8.8)
- stir bar

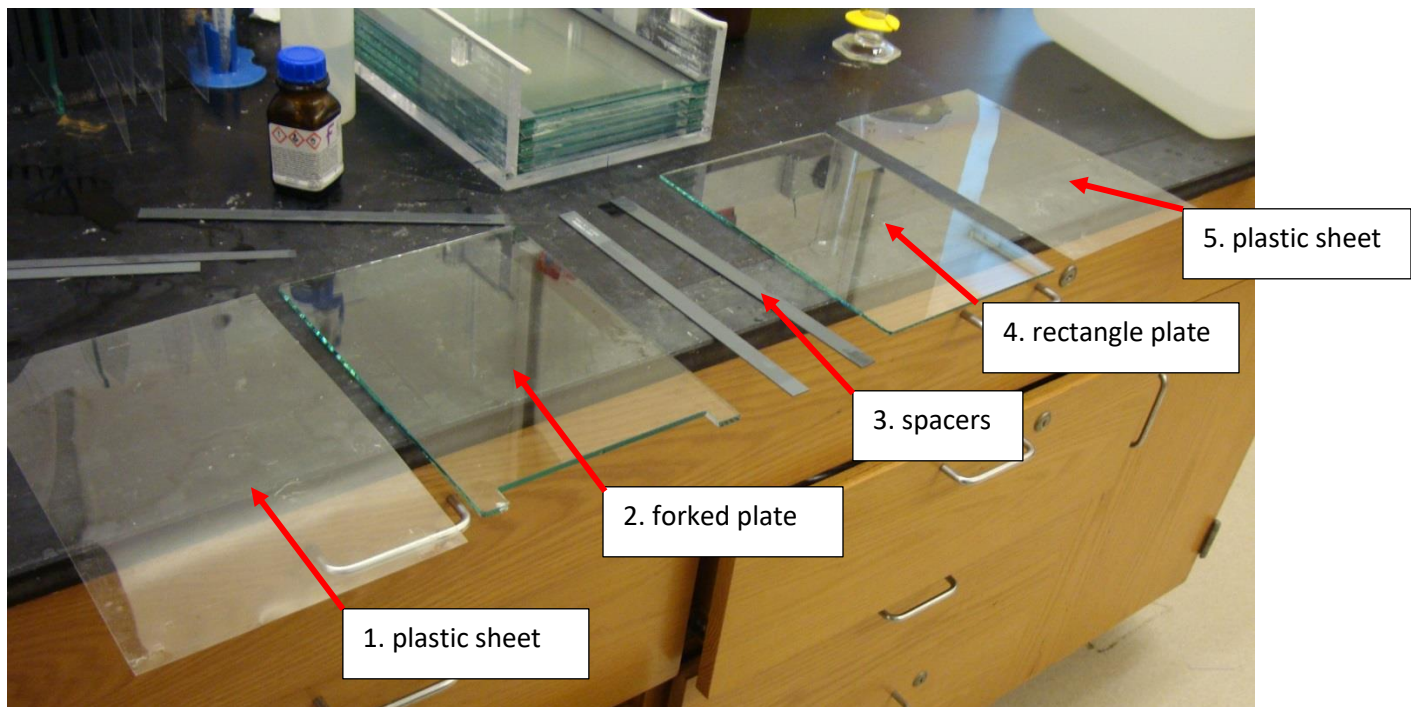
4. make sure stir bar is rotating and de-gas for ~40-to-60 minutes. (No Heat; no need to turn heater on)



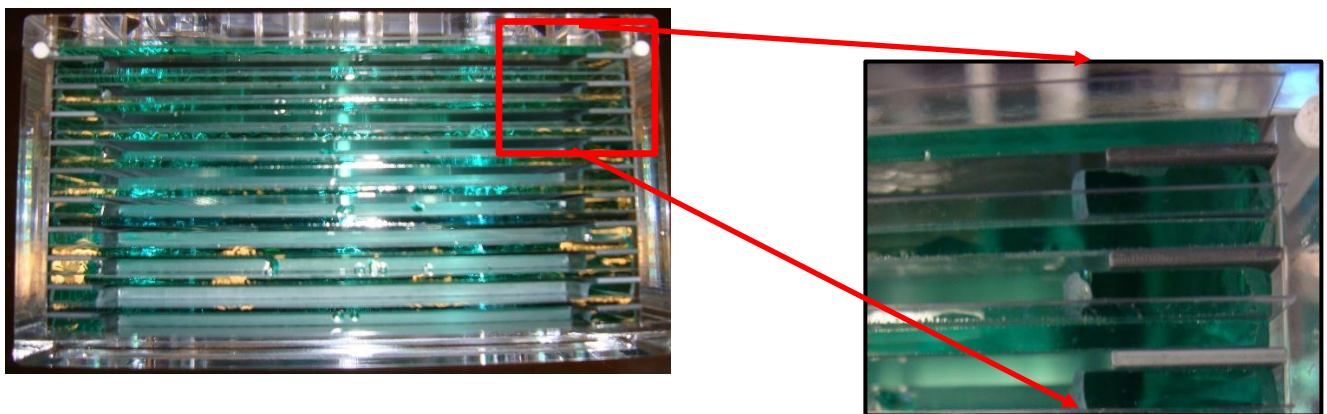
5. While waiting for the mixtue to mix, you can put together the rig. Finger tight the bolt on lid.

ORDER OF ITEMS WHEN MAKING RIG:

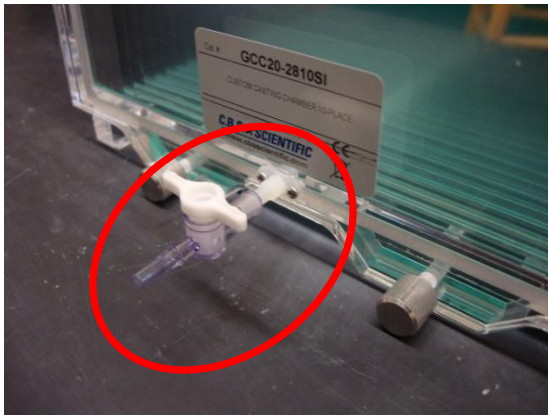
- 1-plastic sheet
- 2-forked plate
- 3-spacers
- 4-rectangle plate
- 5- repeat steps 1 through 4: for a total of 9 times (total of 10 plates in rig)
- 6-last item is a plastic sheet before closing



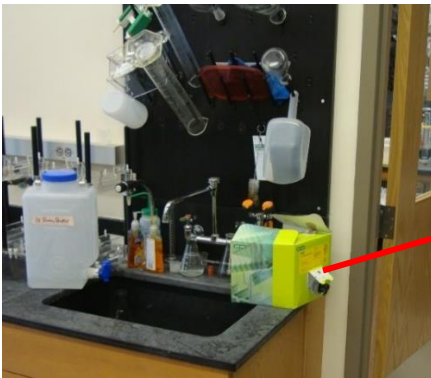
6. Make sure stacks are correctly placed, by observing at the symmetry/uniformity of the glass.



7. make sure plug is inserted at the bottom on rig.



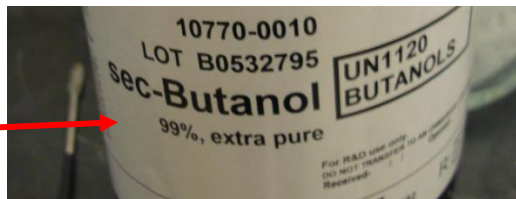
8. Make 10% SDS solution. Take 1mL from the 10X carton, as shown below.
(1mL, 1x SDS, 9mL ddiH₂O).



9. Make APS solution (550 mg APS in 5.5 mL ddiH₂O). (Make APS fresh solution) Make sure ddiH₂O reaches and does **not go beyond 5.5mL on the conical tube.**



10. Have ready: Place sec-butanol on the countertop (Float 750μL butanol over top of each glass plate in the cassette).



11. After ~40-to-60 minutes, remove flask from de-gas connector.



TIMING IS EVERYTHING:

While stirring on the stirrer, ADD the three reagents in this order:

-1- 7.2mL 10% SDS

-2- 3.6mL APS solution

FINALLY ADD (TEMEM is the LAST reagent add):

-3- 360 μ L TEMEM as the final reagent

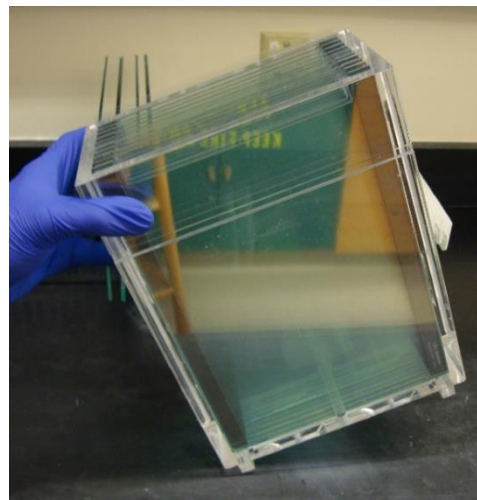
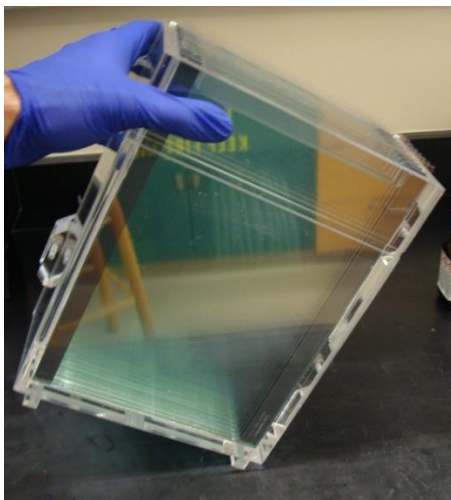
(QUICKLY pour into rig)

a---pour in the middle

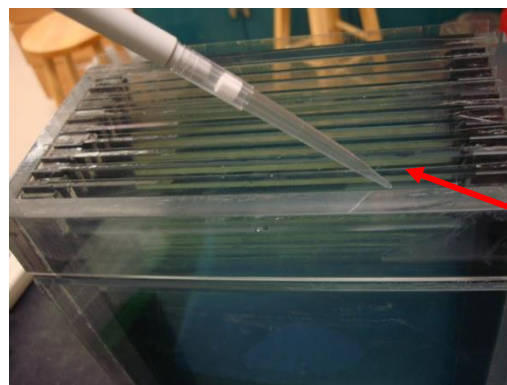
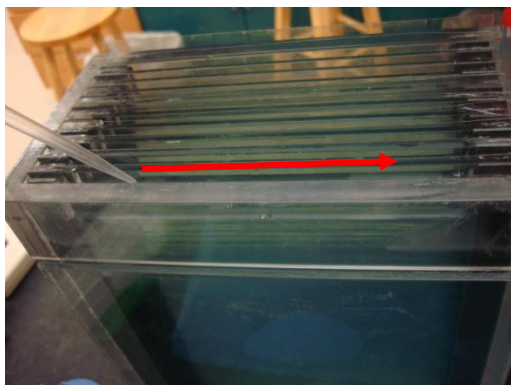
b---pour **ALL** of the mixture (it will "seem" more than needed)

12. After pouring the final reagents:

a---Tilt the rig, to the right and then to the left. Drop the rig when tilted, should hear a loud "bang."

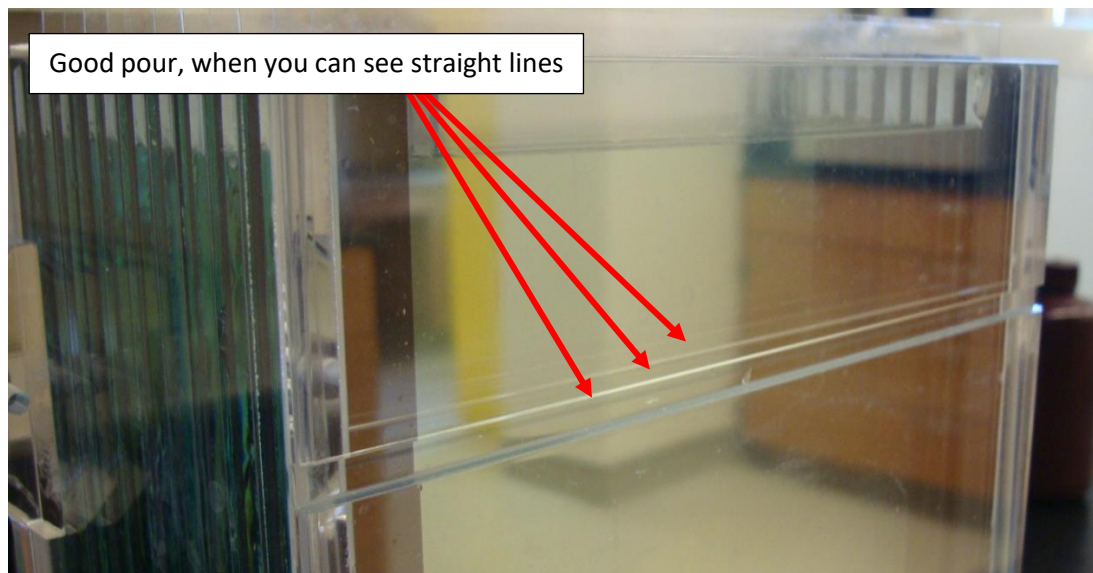


13. Begin pipetting the 750 μ L butanol over top of each section ~20cm from the edge until you reach ~20cm from the other edge, as shown below.

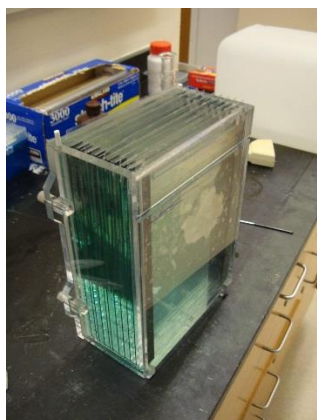


stop prior to
the edge, as
shown

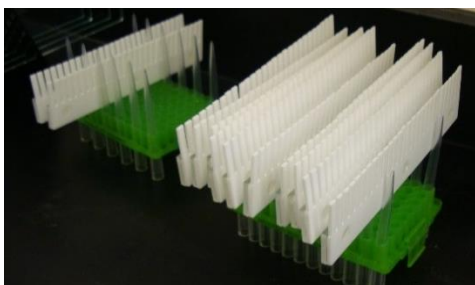
14. --You will be able to verify your technique, if you see straight lines.



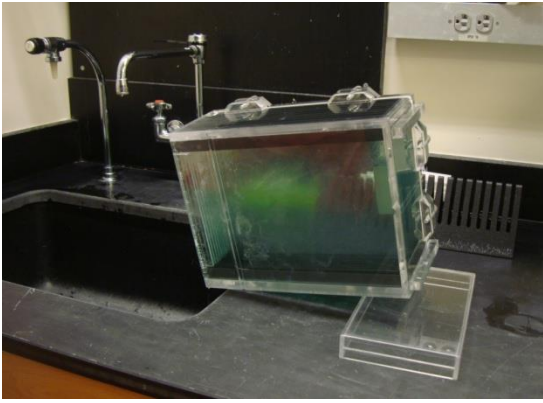
15. Let sit for ~45-to-60 minutes, until extra gel solution in the flask has completely polymerized.



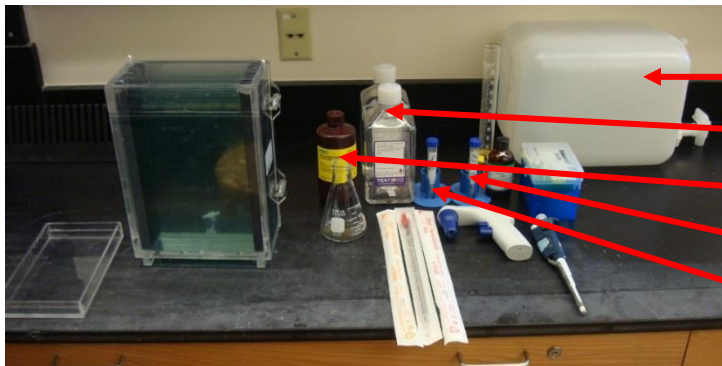
16. Have 10 dry combs out. They should be clean and dry. Also the combs should be level and even.



17. Once ~45 minutes have passed, pour off butanol, give it a few minutes for excess butanol to evaporate



18. Make stacking gel into a 200mL flask



- 1- 136 mL ddH₂O
- 2- 24 mL 1.5M Tris (6.8)
- 3- 25mL 30% acrylamide solution
- 4- 1.9mL 10% SDS
- 5- 938μL APS solution

FINALLY ADD (TEMEM is the LAST reagent add):

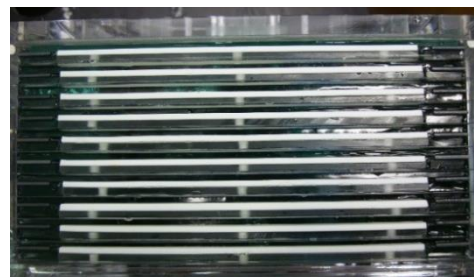
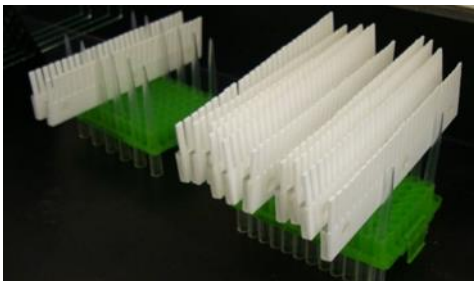
--188μL TEMEM as the final reagent

(QUICKLY pour into rig)

a---pour on each well evenly

b---pour **ALL** of the mixture

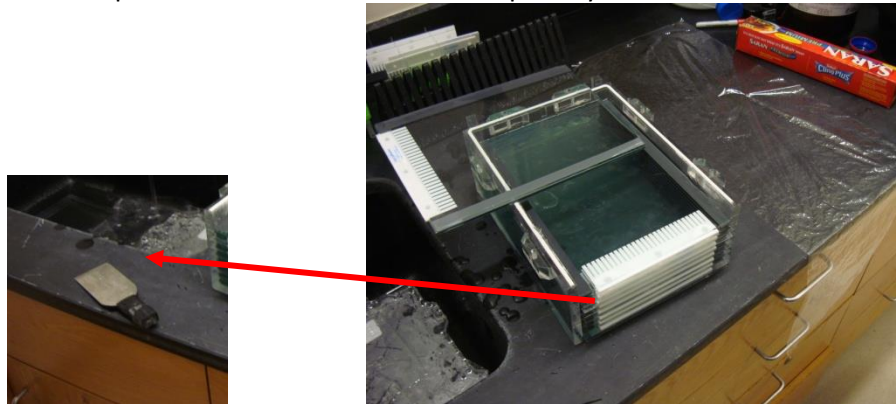
19. After pouring, quickly place combs in each of the rigs. Rig should look symmetrical and uniform.



20. Place rig in cold room (4°C refrigerator) overnight. Make sure lid is placed over the rig.



21. Use metal spatula to separate cassettes. Have Saran wrap ready.



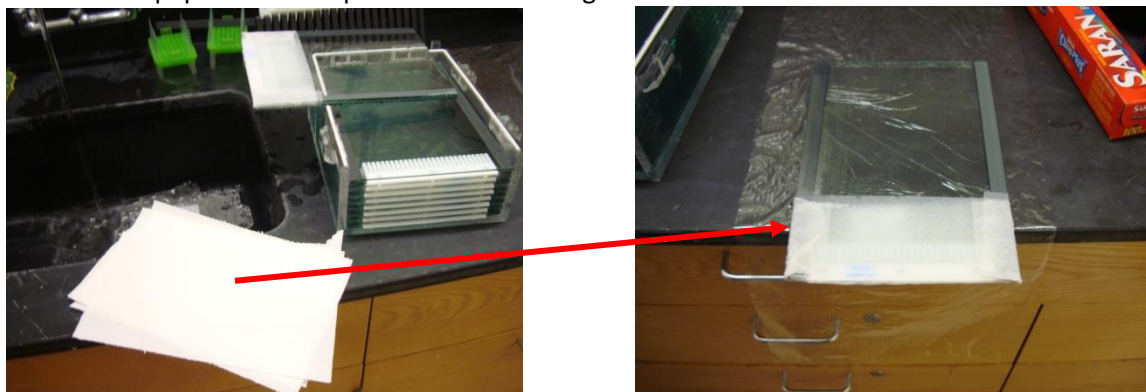
22. Following day, Inspect gel. This is an image of a bad gel. Take a look at the red circles which indicate a bad sections on the gel. Discard gel



23. Inspect gel. This is a good gel. One should see a straight line. Keep this gel.



24. Use one paper towel stripe to cover comb region of the cassette.



25. Mark your initials and date. Then, place all finished gel cassette in the 4°C refrigerator.



Revised: August 03, 2022
Last modified by: Roberto Mendez